Math Refresher for Masters

Homework 6

1 $$\int_0^2 x e^{x^2} dx = \int_0^2 \frac{e^{x^2}}{2} dx^2 = \frac{e^{x^2}}{2}\Big|_0^2 = \frac{1}{2}\left(e^{2^2} - e^{0^2}\right) = \boxed{\frac{1}{2}e^4 - \frac{1}{2}}$$

2 $$\int_0^1 e^x \cos x\, dx = \begin{bmatrix} u = e^x & u' = e^x \\ v' = \cos x & v = \sin x \end{bmatrix} = e^x \cdot \sin x\Big|_0^1 - \int_0^1 e^x \cdot \sin x\, dx = \begin{bmatrix} u = e^x & u' = e^x \\ v' = \sin x & v = -\cos x \end{bmatrix} =$$

$$= e^x \cdot \sin x\Big|_0^1 + e^x \cdot \cos x\Big|_0^1 - \int_0^1 e^x \cdot \cos x\, dx \quad \Rightarrow \int_0^1 e^x \cos x\, dx = \frac{1}{2} e^x (\sin x + \cos x)\Big|_0^1 =$$

$$= \frac{1}{2} e (\sin 1 + \cos 1) - \frac{1}{2} \cdot 1 \cdot (\sin 0 + \cos 0) = \boxed{\frac{e(\sin 1 + \cos x) - 1}{2}}$$

3 $$\int \sin x \cdot \cos x\, dx = \int \sin x\, d(\sin x) = \boxed{\frac{\sin^2 x}{2} + C}$$

4 $$\int x^2 \cdot \cos x\, dx = \begin{bmatrix} u = x^2 & u' = 2x \\ v' = \cos x & v = \sin x \end{bmatrix} = x^2 \cdot \sin x - \int 2x \sin x\, dx = \begin{bmatrix} u = x & u' = 1 \\ v' = \sin x & v = -\cos x \end{bmatrix} =$$

$$= x^2 \cdot \sin x + 2x \cdot \cos x - 2\int \cos x\, dx = \boxed{x^2 \cdot \sin x + 2x \cdot \cos x - 2\sin x + C}$$

5 $$\int_0^2 \frac{2x}{x^2+4} dx = \int_0^2 \frac{1}{x^2+4} d(x^2) = \ln(x^2+4)\Big|_0^2 = \ln 8 - \ln 4 = \ln \frac{8}{4} = \boxed{\ln 2}$$

6 $x \in [0; 1]$

$$h(x) = \begin{cases} x, & x \neq \frac{1}{2} \\ 10, & x = \frac{1}{2} \end{cases}$$

$$\lim_{x \to \frac{1}{2}} h(x) = \lim_{x \to \frac{1}{2}} x = \frac{1}{2} \neq h\left(\frac{1}{2}\right) = 10 \quad \Rightarrow \quad \exists a : \lim_{x \to a} h(x) \neq h(a) \Rightarrow \boxed{h(x) \text{ is not continuous at } x = \frac{1}{2}}$$

This function is Riemann integrable, because it is bounded and continuous almost everywhere.

$$\int_0^1 h(x) dx = \int_0^1 x\, dx = \frac{x^2}{2}\Big|_0^1 = \boxed{\frac{1}{2}}$$

7 $$F(x) = \int_0^{x^2} e^{-t^2} dt$$ Fundamental Theorem of Calculus: $F(x) = \int_a^x f(t) dt \Rightarrow F'(x) = f(x)$

$$F'(x) = \frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x)) \cdot g'(x), \quad f(t) = e^{-t^2}, \quad g(x) = x^2, \quad g' = 2x \Rightarrow \boxed{F'(x) = e^{-x^4} \cdot 2x}$$